

Smart Factory Environmental Monitoring Gateway

Technical Specification and Verification Basis

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This synthetic but realistic specification is intentionally structured to test document ingestion, requirement extraction, traceability, chatbot grounding, risk analysis, and generated test plan quality. It contains explicit requirement identifiers, acceptance criteria, interfaces, operating modes, exceptions, and open issues.

1. Purpose and Scope

The Smart Factory Environmental Monitoring Gateway (SF-EMG) is an edge device and cloud-connected service used to monitor temperature, humidity, particulate matter, vibration, and volatile organic compound levels in a medium-sized production facility. The system raises local alarms, forwards validated telemetry to the factory cloud, and maintains an auditable data trail for quality and safety investigations.

This document defines the product behavior that must be verified before pilot deployment on Assembly Line B. It is not a user manual. It is the primary source for system, integration, security, and acceptance testing.

Out of scope: procurement of sensors, design of the cloud analytics dashboard beyond the ingestion API, and certification testing performed by external laboratories.

1.1 Product Context

Item	Definition
Deployment site	Indoor production area with five sensor zones, one supervisor console, and a secure factory network.
Main users	Maintenance engineer, quality engineer, shift supervisor, IT administrator, and external auditor.
Critical outcome	Environmental anomalies must be detected and communicated within controlled time limits without creating false audit records.
Pilot duration	Eight weeks, with continuous telemetry capture and weekly export of compliance reports.

2. System Overview

The SF-EMG platform consists of four logical subsystems: sensor acquisition, edge validation and buffering, local operator interface, and cloud synchronization. The edge gateway runs a Linux-based runtime and communicates with sensors over RS-485 Modbus and BLE. Cloud synchronization uses MQTT over TLS 1.3.

Data is processed in the following sequence: raw sensor frame, timestamp normalization, plausibility validation, calibrated engineering value, zone aggregation, alarm rule evaluation, local persistence, and cloud publication. Every step after validation must preserve the source sensor identifier and original frame timestamp.

2.1 Architecture Components

Component ID	Component	Responsibility	Verification focus
CMP-01	Sensor Adapter Service	Polls wired and wireless sensors, retries transient failures, records acquisition quality.	Protocol handling, timeout behavior, bad-frame rejection.
CMP-02	Validation Engine	Applies calibration offsets, range limits, drift rules, and data quality flags.	Boundary values, calibration expiration, invalid data propagation.
CMP-03	Alarm Manager	Evaluates configurable alarm rules per zone and drives local outputs.	Alarm latency, hysteresis, acknowledgement, escalation.
CMP-04	Local Store	Stores telemetry, configuration, audit entries, and unsent cloud events.	Retention, crash recovery, export integrity.
CMP-05	Cloud Sync Agent	Publishes telemetry and audit events using MQTT over TLS.	Offline buffering, reconnect, duplicate suppression.
CMP-06	Operator Web UI	Provides status, alarms, calibration notices, exports, and admin screens.	Role access, refresh behavior, accessibility, input validation.

3. Functional Requirements

Req ID	Requirement	Acceptance criteria	Priority
FR-001	The gateway shall poll every enabled wired sensor at least once every 10 seconds during Normal mode.	In a 30-minute run with 20 sensors, at least 99.5 percent of expected polls are completed or recorded with a quality failure reason.	Must
FR-002	The gateway shall accept BLE sensor advertisements only from registered sensor IDs assigned to the active project.	Unregistered BLE frames are ignored, logged as security events, and never displayed as valid telemetry.	Must

Req ID	Requirement	Acceptance criteria	Priority
FR-003	The validation engine shall apply the latest approved calibration offset before alarm rule evaluation.	A test calibration offset changes the displayed value and alarm decision within the next acquisition cycle.	Must
FR-004	The alarm manager shall raise a High Temperature alarm when a zone temperature remains above 38.0 C for 60 consecutive seconds.	Alarm activates between 60 and 75 seconds after threshold crossing and cites the triggering zone and sensor IDs.	Must
FR-005	The alarm manager shall clear a High Temperature alarm only after the zone temperature remains below 36.5 C for 120 consecutive seconds.	Alarm does not flap when temperature oscillates between 36.6 C and 38.1 C.	Must
FR-006	Operators shall acknowledge active alarms without deleting the alarm history.	Acknowledgement adds user, timestamp, and comment; original raise/clear events remain immutable.	Must
FR-007	The gateway shall buffer cloud-bound telemetry and audit events while the network is unavailable.	At least 72 hours of pilot-site telemetry at 20 sensors and 10-second cadence is retained without data loss.	Must
FR-008	After cloud reconnection, buffered events shall be sent in original event-time order per sensor.	Published sequence for a sensor is monotonic by event timestamp, excluding explicitly marked duplicates.	Should
FR-009	The operator UI shall show per-zone environmental status using Fresh, Warning, Alarm, and Sensor Fault states.	State changes appear in the UI within 5 seconds of backend state change on a stable LAN.	Must
FR-010	The system shall export a signed CSV and JSON audit bundle for a selectable date range.	Export includes telemetry hashes, alarm history, configuration changes, user acknowledgements, and signature metadata.	Must
FR-011	The system shall reject configuration changes that would disable all sensors in an active zone.	Attempt is rejected with a visible validation message and audit event.	Must
FR-012	Administrators shall be able to import sensor-zone mappings from a CSV file.	Valid CSV creates mappings; invalid rows are reported with row numbers and do not partially corrupt existing mappings.	Should

4. Non-Functional and Performance Requirements

Req ID	Category	Requirement	Measure
NFR-001	Availability	The edge gateway shall continue local monitoring when the cloud is unavailable.	All local alarms and UI status remain functional for 72 hours offline.
NFR-002	Latency	Critical alarm publication to the local UI shall complete quickly enough for operator response.	P95 from validated input sample to visible alarm state is <= 3 seconds.
NFR-003	Durability	The local store shall survive sudden power loss without losing committed audit events.	After 50 forced power cycles, no committed audit event is missing or malformed.
NFR-004	Capacity	The gateway shall support at least 40 sensors and 10 operator sessions.	No API endpoint exceeds 1.5 seconds P95 under defined load.
NFR-005	Maintainability	Configuration changes shall be reversible through versioned configuration snapshots.	Rollback restores previous active configuration and emits audit event.
NFR-006	Observability	The system shall expose health, readiness, and metric endpoints for IT monitoring.	Endpoints report database, sensor adapter, queue, and cloud sync health.

5. Security and Roles

Authentication uses corporate SSO in production and local administrator accounts in isolated pilot mode. All administrative actions must be attributable to a user. Shared accounts are prohibited for production pilots.

Req ID	Security requirement	Evidence expected
SEC-001	Users shall authenticate before viewing telemetry, alarm history, or exports.	Unauthenticated requests are denied and no sensitive payload is returned.
SEC-002	Only Admin users shall create users, edit role assignments, configure MQTT credentials, or import sensor mappings.	Non-admin attempts produce HTTP 403 and audit event.

Req ID	Security requirement	Evidence expected
SEC-003	Passwords for local pilot accounts shall be stored using a memory-hard password hash.	Database inspection shows no plaintext or reversible password storage.
SEC-004	MQTT communication shall use TLS 1.3 with server certificate validation enabled.	Connection to a broker with invalid certificate fails closed.
SEC-005	Audit exports shall not include secret values such as API tokens, MQTT passwords, private keys, or session tokens.	Export review and automated secret scan find no secret values.
SEC-006	Security-relevant failures shall be rate-limited to avoid log flooding.	Repeated invalid login attempts produce bounded log volume and account lockout after policy threshold.

6. Interfaces

The gateway exposes a local HTTPS API for the operator UI and publishes cloud telemetry through MQTT. The public API is not intended for direct operator scripting during the pilot, but must be stable enough for monitoring and integration tests.

6.1 MQTT Topic Contract

Topic	Direction	Payload summary	QoS
factory/{siteId}/gateway/{gatewayId}/telemetry	Publish	Validated sensor readings with sensorId, zoneId, value, unit, quality, eventTime, receivedTime, calibrationVersion.	1
factory/{siteId}/gateway/{gatewayId}/alarm	Publish	Alarm raise, acknowledge, clear, and escalate events with immutable alarmId.	1
factory/{siteId}/gateway/{gatewayId}/command	Subscribe	Cloud command envelope for export request, config sync, and diagnostics trigger.	1
factory/{siteId}/gateway/{gatewayId}/health	Publish	Gateway software version, storage pressure, queue depth, clock sync status, and sensor adapter status.	0

6.2 Local REST API Examples

Endpoint	Method	Purpose	Important behavior
/api/zones/{zoneId}/status	GET	Read current zone status.	Returns state, active alarms, latest reading summary, and data freshness.
/api/alarms/{alarmId}/acknowledge	POST	Acknowledge an alarm.	Requires Operator or Admin role; comment length 5 to 500 characters.
/api/config/sensors/import	POST	Import sensor mapping CSV.	Requires Admin; validates entire file before applying changes.
/api/exports/audit-bundle	POST	Create signed export.	Requires Quality Engineer or Admin; asynchronous job returns exportId.
/readyz	GET	Readiness probe.	Reports dependency readiness for local store, queue, cloud sync, and configuration validity.

7. Operating Modes and State Rules

Mode	Entry condition	Allowed behavior	Exit condition
Commissioning	First boot or Admin selects commissioning.	Sensor mapping, calibration import, network setup, simulated alarms allowed.	All mandatory sensors mapped and gateway activated.
Normal	Gateway activated with valid configuration.	Continuous polling, validation, alarms, buffering, sync, exports.	Admin enters Maintenance or system detects critical fault.
Maintenance	Admin starts maintenance window.	Sensor disablement allowed per zone except all sensors in active zone; alarms marked maintenance-suppressed.	Admin ends window or scheduled end time reached.

Mode	Entry condition	Allowed behavior	Exit condition
Degraded	Sensor adapter, storage pressure, or network problem exceeds threshold.	Local alarms continue where possible; cloud sync may pause; UI shows degraded reason.	Dependency returns healthy and backlog is below threshold.
Safe Shutdown	Power-loss signal or critical storage failure.	Stop new config changes; flush audit log; close local store.	Manual restart after condition resolved.

State conflict rule: Alarm state has priority over Warning, and Sensor Fault has priority over Fresh but not over active Alarm for the same zone. If all sensors in a zone are stale for more than 90 seconds, the zone state shall be Sensor Fault unless an uncleared safety alarm is already active.

8. Data and Retention Rules

- Telemetry records are immutable after validation. Corrections must be added as new records with a correctionOf reference.
- Audit events are retained locally for at least 400 days or until successful signed export plus administrator-approved purge.
- Raw sensor frames are retained for 14 days during pilot mode and may be disabled after pilot approval.
- Configuration snapshots are retained for every activation, import, rollback, or role-permission change.
- Export signatures use SHA-256 over canonical JSON payloads. CSV files include the same record identifiers as JSON exports.

9. Verification Strategy

Testing shall combine automated unit tests, integration tests with simulated sensors, system tests on the pilot hardware, security tests, and user acceptance tests with shift supervisors. Generated test plans must preserve requirement IDs and should identify which tests require hardware, cloud broker access, or seeded historical data.

Test level	Scope	Required assets
Unit	Validation rules, threshold calculations, role checks, payload schema validation.	Mock time source, calibration fixtures, invalid frame examples.
Integration	Sensor adapter with simulator, local store, alarm manager, REST API, MQTT client.	Modbus simulator, BLE advertisement replay, local broker, seeded DB.
System	Full gateway with pilot network, operator UI, offline buffering, exports.	Hardware gateway, test sensors, factory VLAN, cloud staging broker.
Security	Authentication, authorization, secret handling, TLS validation, audit integrity.	Test accounts, invalid certificates, token scanner, audit export samples.
Acceptance	Representative operator workflows and weekly quality export review.	Shift supervisor checklist, pilot scenario script, acceptance sign-off form.

9.1 Requirement Traceability Matrix

Requirement IDs	Recommended test objective	Primary evidence
FR-001, FR-002	Verify polling cadence, registration filtering, and acquisition quality recording.	Sensor simulator logs, gateway event log, rejected BLE event audit.
FR-003, FR-004, FR-005	Verify calibrated values drive alarm raise/clear rules with hysteresis and timing bounds.	Time-series fixture, UI screenshots, alarm event records.
FR-006, SEC-001, SEC-002	Verify authenticated role-based alarm acknowledgement without history mutation.	HTTP responses, DB audit rows, UI role behavior.
FR-007, FR-008, NFR-001	Verify offline buffering capacity and ordered replay after reconnection.	Network outage run log, MQTT broker capture, backlog depth metrics.
FR-010, SEC-005, NFR-003	Verify signed export completeness, secret exclusion, and durable committed audit events.	Generated export bundle, hash verification, secret scan output.
FR-011, FR-012, NFR-005	Verify CSV import validation, all-or-nothing application, and rollback.	Invalid CSV cases, configuration snapshots, rollback audit trail.

Requirement IDs	Recommended test objective	Primary evidence
NFR-002, NFR-004, NFR-006	Verify latency, capacity, and observability endpoint behavior under load.	Load-test report, metrics scrape, readiness response samples.
SEC-003, SEC-004, SEC-006	Verify password hashing, TLS failure behavior, and abuse-rate controls.	DB inspection, TLS negative test logs, rate-limit evidence.

10. Acceptance Scenarios

Scenario ID	Narrative	Expected outcome
AS-001	A zone gradually heats above 38.0 C, remains high for 70 seconds, then cools below 36.5 C for two minutes.	High Temperature alarm raises, can be acknowledged, and clears only after the low threshold duration.
AS-002	Factory network is disconnected for 90 minutes while sensors continue publishing locally.	Local UI and alarms continue; cloud backlog grows; reconnect publishes ordered events without duplicates.
AS-003	A maintenance engineer imports a CSV where row 14 references an unknown zone.	Import is rejected before applying any rows; the error includes row 14 and existing mappings remain active.
AS-004	A non-admin shift supervisor attempts to edit MQTT credentials.	Access is denied; no credential is changed; security audit event records the denied action.
AS-005	An auditor requests a signed export for the previous week.	Export completes asynchronously, includes telemetry and audit evidence, excludes secrets, and signature validates.

11. Known Ambiguities and Open Decisions

The following open points are intentionally included to test whether the AI platform can separate explicit requirements from unresolved decisions.

Open ID	Issue	Impact if unresolved
OD-001	The final cloud broker certificate rotation procedure is not defined.	Security and maintenance test cases cannot be fully closed.
OD-002	The pilot team has not decided whether VOC alarms are advisory or safety-critical.	Alarm priority matrix and notification escalation tests may change.
OD-003	CSV import format allows optional buildingId, but the cloud topic hierarchy already includes siteId only.	Integration mapping tests need final field ownership decision.
OD-004	The exact account lockout threshold is owned by corporate IT and not yet available.	SEC-006 can only be tested with placeholder policy until IT confirms.

12. Change History

Version	Date	Author	Summary
1.1	2026-03-15	Systems Engineering	Initial pilot behavior, sensor polling, and cloud sync scope.
1.2	2026-04-08	Quality Engineering	Added signed export and audit retention requirements.
1.3	2026-05-02	Security Review	Added role permissions, TLS validation, and secret exclusion requirements.
1.4	2026-06-12	Pilot Readiness Board	Added acceptance scenarios, open decisions, and readiness endpoint requirement.

Appendix A. Test Data Hints

To exercise generated test instructions, use the following sample values: registered BLE sensor BLE-A17 assigned to Zone Z-PAINT-02; unregistered BLE sensor BLE-UNKNOWN-9; wired Modbus sensor MB-03 with calibration offset +1.8 C; MQTT gatewayId GW-LINEB-01; siteId SITE-CASA-01. Boundary temperature values: 36.4 C, 36.5 C, 37.9 C, 38.0 C, 38.1 C. Alarm timing boundaries: 59 seconds, 60 seconds, 75 seconds, 119 seconds, and 120 seconds.

Appendix B. Example Payloads and Glossary

This appendix gives concrete payload examples that can be used to test whether generated instructions include realistic API/MQTT data and expected assertions.

B.1 Valid MQTT Telemetry Payload

```
{ "siteId": "SITE-CASA-01", "gatewayId": "GW-LINEB-01", "sensorId": "MB-03", "zoneId": "Z-PAINT-02", "metric": "temperature", "value": 38.1, "unit": "C", "quality": "validated", "eventTime": "2026-06-12T10:15:30Z", "receivedTime": "2026-06-12T10:15:31Z", "calibrationVersion": "CAL-2026-05-A" }
```

B.2 Invalid Sensor Mapping CSV Rows

Row	CSV content	Expected validation result
14	BLE-A17,Z-UNKNOWN,temperature,enabled	Reject entire import because zone does not exist.
18	MB-03,Z-PAINT-02,temperature,enabled	Accept only if MB-03 is not already mapped to another active zone.
23	BLE-UNKNOWN-9,Z-PACK-01,humidity,enabled	Reject because sensor is not in the approved device registry.

B.3 Negative Security Test Seeds

Seed ID	Input	Expected behavior
NST-001	Supervisor account calls POST /api/config/sensors/import.	HTTP 403, no mapping change, security audit event.
NST-002	MQTT broker presents expired certificate.	Connection fails closed; no telemetry is published over insecure channel.
NST-003	Audit export date range contains no telemetry records.	Export succeeds with empty telemetry section, full metadata, and valid signature.
NST-004	Login password is wrong 10 consecutive times.	Policy-controlled lockout or rate limit; no stack traces or secret values in response.

B.4 Glossary

Term	Meaning
Calibration offset	Approved numeric adjustment applied to a raw sensor value before alarm evaluation.
Event-time order	Ordering by the original timestamp of the validated event, not by the time it was replayed after reconnection.
Fresh state	Zone state indicating recent valid readings and no warning, alarm, or sensor fault condition.
Signed audit bundle	Export package containing canonical JSON, CSV projections, hashes, and signature metadata for review.
Sensor Fault	Zone state indicating stale, missing, invalid, or untrusted readings.